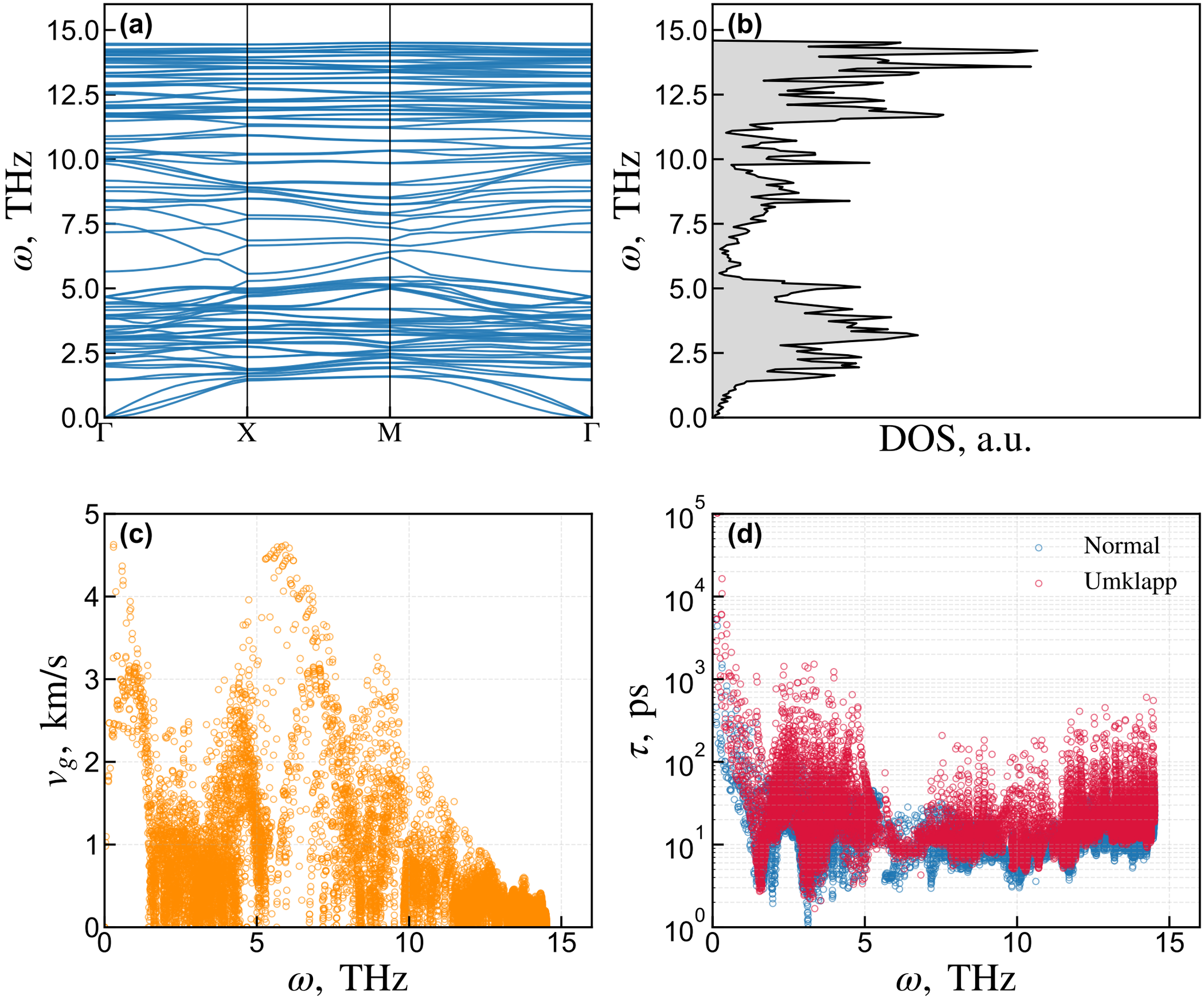
(a)
15.0
12.5
10.0
7.5
5.0
2.5
0.0
ω, THz
Γ
X
M
Γ
(b)
DOS, a.u.
(c)
v_g, km/s
ω, THz
(d)
Normal
Umklapp
τ, ps
ω, THz